defined in `AlarmActions` are executed: in this case, the recovery of the virtual machine—a built-in action for EC2 instances.

In summary, AWS checks the status of the virtual machine every minute. The result of these checks is written to the `StatusCheckFailed_System` metric. The alarm checks this metric, and if there are five consecutive failed checks, the alarm trips.

Listing 13.2 Creating a CloudWatch alarm to recover a failed EC2 instance

```
RecoveryAlarm:                                                    ①
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmDescription: 'Recover EC2 instance when underlying hardware fails.'
    Namespace: 'AWS/EC2'                                          ②
    MetricName: 'StatusCheckFailed_System'                        ③
    Statistic: Maximum                                            ④
    Period: 60                                                    ⑤
    EvaluationPeriods: 5                                          ⑥
    ComparisonOperator: GreaterThanThreshold                      ⑦
    Threshold: 0                                                  ⑧
    AlarmActions:                                                 ⑨
    - !Sub 'arn:aws:automate:${AWS::Region}:ec2:recover'
    Dimensions:
    - Name: InstanceId                                            ⑩
      Value: !Ref Instance
```

① Creates a CloudWatch alarm to monitor the health of the virtual machine

② The metric to monitor is provided by the EC2 service with the namespace AWS/EC2.

③ The name of the metric

④ Statistical function to apply to the metric. The minimum is to notify you if a single status check failed.

⑤ Duration for which the statistical function is applied, in seconds. Must be a multiple of 60.

⑥ Number of time periods over which data is compared to the threshold

⑦ Operator for comparing the output of the statistical function with the threshold

⑧ Threshold triggering an alarm

⑨ Action to perform in case of an alarm. Uses the predefined recovery action for EC2 instances.

⑩ The virtual machine is a dimension of the metric.

While you have been reading through the details, the CloudFormation stack should have reached the status `COMPLETE`. Run the following command to get the output of the stack. If the output is `null`, retry after a few more minutes: